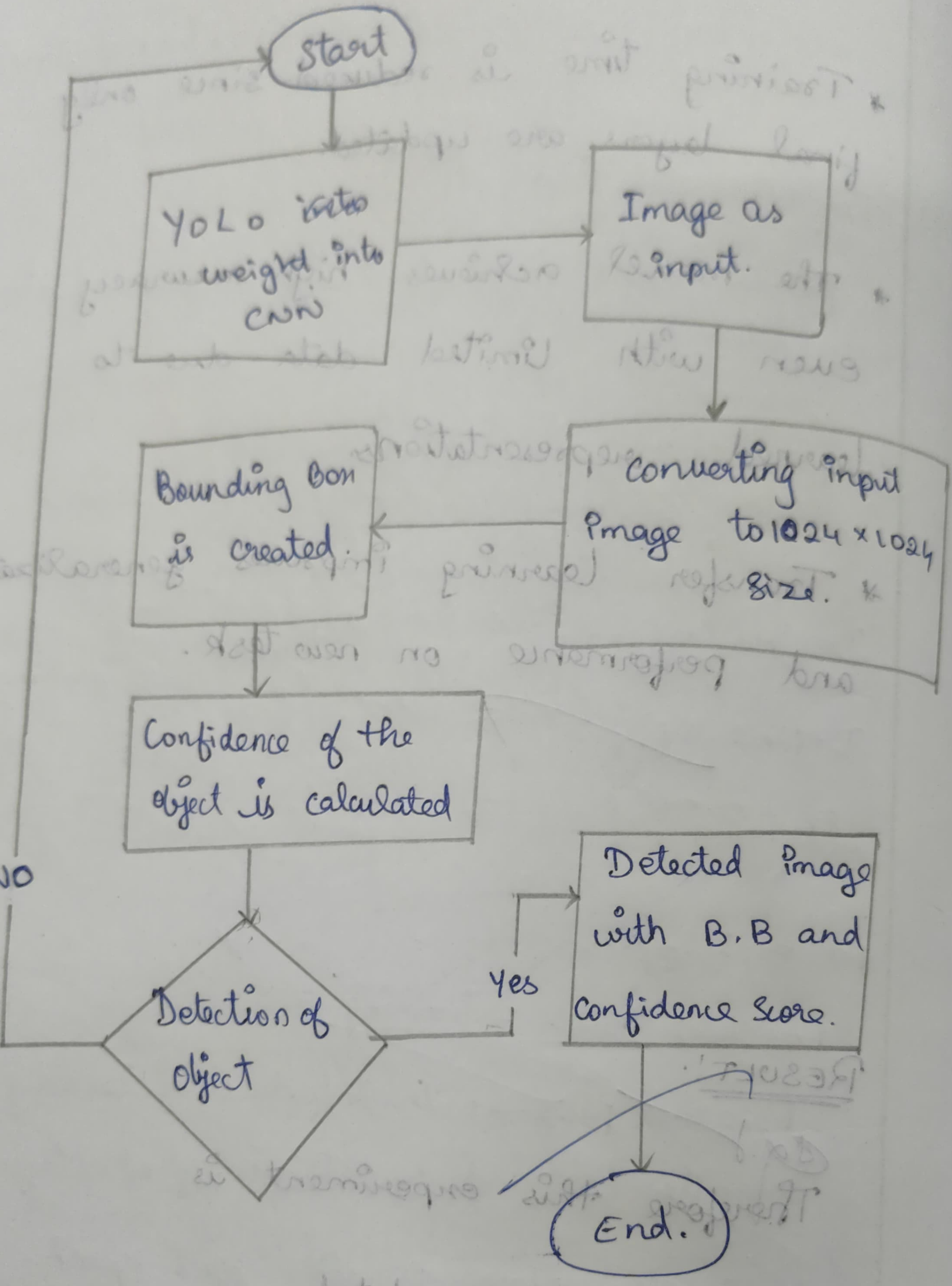
Architecture:



15. IMPLEMENT A YOLO MODEL TO DETECT OBJECT.

AIM:

To implement a ~~Youne~~ YOLO Model to Detect object.

OBJECTIVE:

* Load a pre-trained YOLO model (eg. YOLO v5 | YOLO v8).

* Detect and classify multiple objects in an image or video frame.

* Draw bounding boxes and labels around detected objects.

* Evaluate model performance based on accuracy and speed.

PSEUDOCODE:

Import necessary libraries (torch, cv2, ultralytics)

Load pre-trained YOLO model:

model = YOLO('yolov8n.pt') # or yolov5s.pt

Load input image or video

result = model (image)

perform detection

For each detection in results:

Extract bounding box coordinates, label, confidence

Draw bounding box and label on the image

Display or save the output image/video.

OBSERVATION:

* The YOLO model accurately detects multiple objects in a single pass.

* It provides real-time performance with high detection speed.

* Bounding boxes and labels correctly mark objects with confidence scores.

* Performance depends on ~~t~~ model size and input resolution.

RESULT:

Therefore this experiment is successfully completed.